

Cortex-M4 Interrupt Pending Behaviour

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ARMv7-M Architecture Ref Manual

DS12024Rev4 STM32L4Sxxx Datasheet

RM0432 STM32L4+ Technical reference manual

PM0214 STM32 Cortex-M4 Programming Manual

Figures from “The Definitive Guide to ARM® Cortex®-M3 and Cortex®-M4 Processors” by Joseph Yiu

Acknowledgement: Slides adapted from Dr Rajesh Panicker

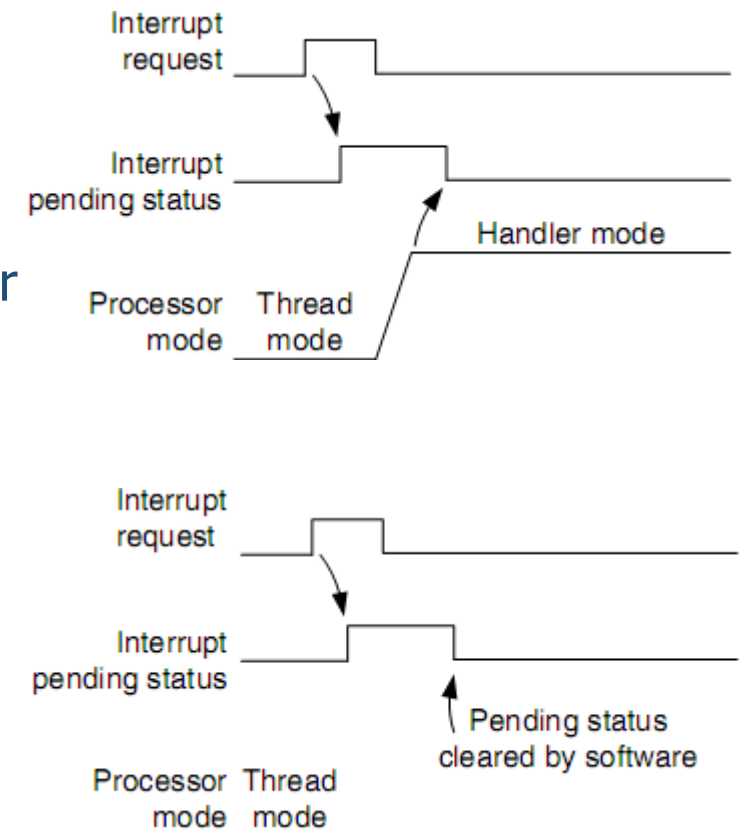


OVERVIEW

- Interrupt Pending Behavior

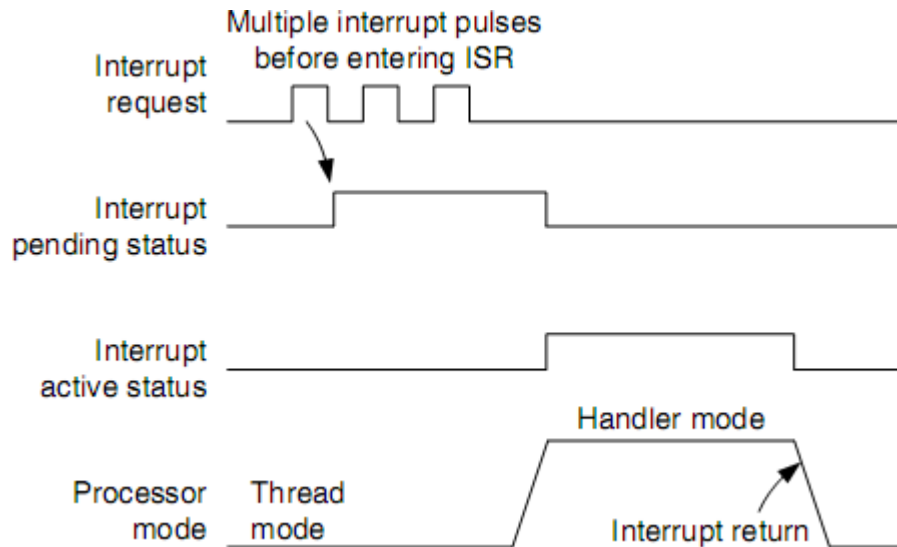
INTERRUPT PENDING BEHAVIOR

- Pending is a state of an exc./int. waiting for the CPU to process
- When an exc./int. is asserted, it will be pended
- Even if the interrupt signal is then de-asserted, pending status is still maintained and causes the interrupt handler to be executed
- When the int. handler starts, pending status will be cleared automatically by the processor/NVIC
- If the int. handler cannot be carried out immediately (e.g., if a higher- priority interrupt handler is running or if the interrupt is disabled), the exc./int. will remain pending
- Pending status can be cleared before the CPU responds to the int. by writing to an interrupt control register in NVIC (not a 'normal' scenario, will see later)

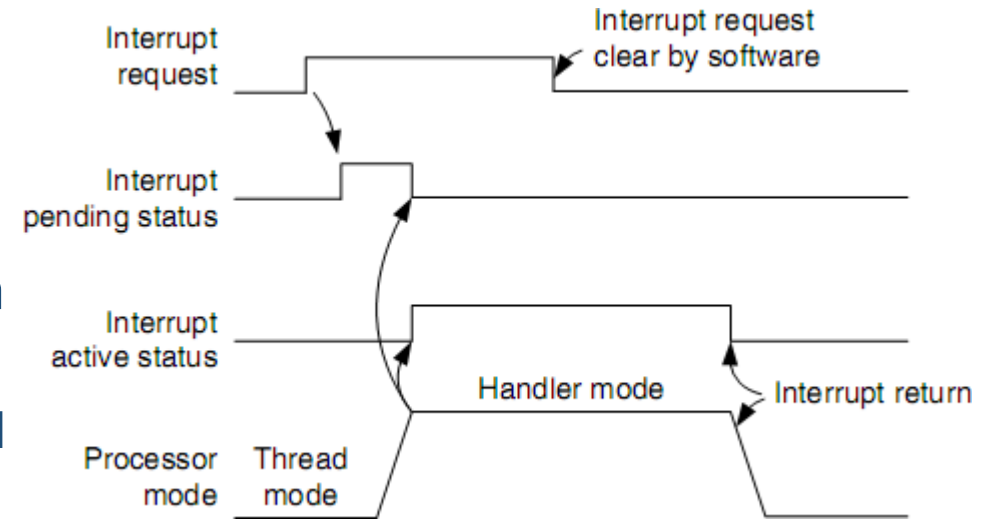


INTERRUPT PENDING BEHAVIOR

- When CPU starts to execute an int., the int. becomes active and pending bit is cleared automatically
- Pending can happen when an int. is disabled. It can trigger the int. sequence when it is enabled later
- Pending -> active happens only if the interrupt is enabled

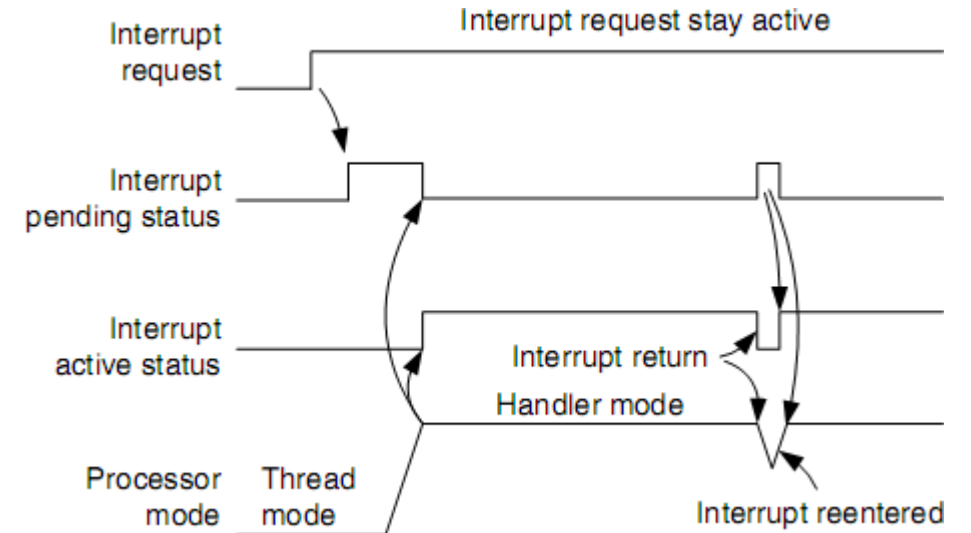
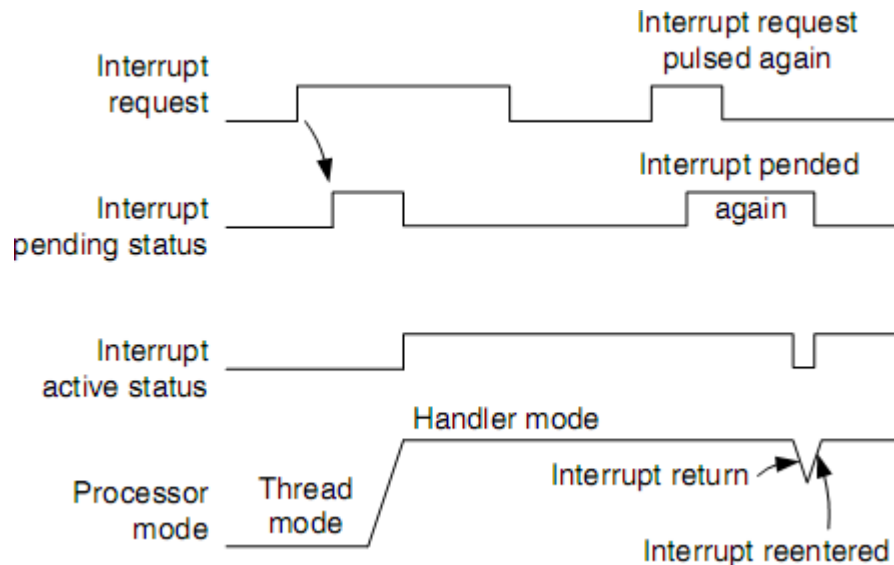


- An int. that is signaled several times before CPU starts to process it will be treated as a single int. request
- Subsequent arrivals of an interrupt which is already pending are 'lost'



INTERRUPT PENDING BEHAVIOR

- If an int. source continues to hold int. request signal active, int. will be pended again at the end of int. service routine (exception handler exit)
 - Typically happens when an interrupt is not 'cleared' by the software



- If a device raises an interrupt while its handler is already running, it will be pended again
 - When an interrupt is raised, it goes pending as long as it is not already pending, irrespective of other factors such as whether it is enabled, whether the corresponding handler is already running



CONCLUSION

- Interrupt Pending Behavior
 - Interrupt Pending status
 - Interrupt Active status
 - Processor mode
 - Thread mode